VeloRad: Modality-Agnostic Radiology Reports Without Autoregression

Anonymous submission

Abstract

The prevailing recipe for radiology report generation couples an ever-larger LLM decoder, now 3 to 8B parameters, to a medical vision encoder. We ask whether structured report generation, whose clinical content is comparatively low-entropy, actually needs this scale. We present **VeloRad**, a continuous flow-matching language model that generates reports by integrating a condition-driven denoising velocity field over a frozen text-encoder latent space, rather than decoding autoregressively. Under a unified RadBERT protocol on chest CT, VeloRad matches the best support-weighted clinical F1 (0.40) among comparable systems using only 0.8B parameters, 3.7 to 10× fewer than the 3 to 8B VLMs it matches, and alone stays balanced across precision and recall. VeloRad's condition interface is moreover *modality-agnostic* and it transfers to 2D chest X-ray with no backbone change and no X-ray-specific vocabulary, reproducing the same behavior we characterize on CT. This means that the clinical evidence is present in the image condition (AUC 0.77 on CT, 0.65 on CXR) and near-perfectly recoverable from ground-truth report latents (0.97/0.82), yet the *generated* latents retain only part of it (0.69/0.62). The information is lost during generation in both modalities. VeloRad thus reframes report generation as evidence-to-text transfer under a compact, modality-agnostic model, with a rarity- and density-stratified evaluation that exposes where aggregate scores hide their failures.

Introduction

Free-text radiology report generation (RRG) from 3D chest CT has advanced rapidly on the back of large vision-language models (VLMs). The dominant recipe pairs a pretrained or fine-tuned medical vision encoder with an autoregressive decoder, scaled to 3-8B parameters. Yet on the standardized CT-RATE benchmark, evaluated by a fixed RadBERT finding extractor, this scaling has not delivered balanced clinical accuracy. The picture is instead one of two failure modes at opposite extremes. Large autoregressive VLMs are *precision-biased to the point of silence*, thus they emit confident text but report only a small fraction of the findings actually present, yielding clinical recall as low as 0.13-0.16. Conversely, recall-oriented encoders *over-predict* and they recover most findings but at a precision near 0.30, flooding the report with false positives. Such a model can even reach an aggregate F1 comparable to ours, but only by operating at a clinically unusable extreme, where a report that flags nearly everything is as untrustworthy as one that flags nothing. Neither operating point is clinically usable.

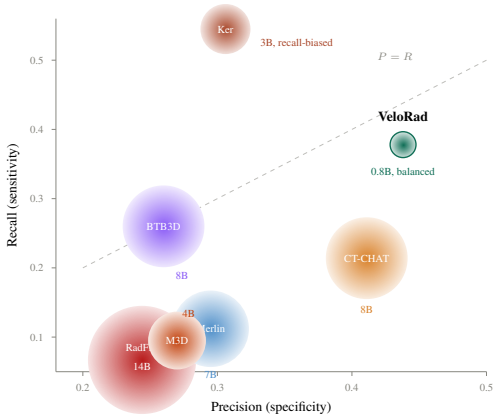


Figure 1: VeloRad attains the highest support-weighted F1 among comparable baselines while remaining close to the balanced $P{=}R$ operating point, using an order of magnitude fewer parameters. Circle area encodes model size; baselines are reproduced under the RadBERT protocol.

We argue that this imbalance, together with the escalating parameter budget, calls the dominant recipe into question. We are investigating if high-quality RRG actually require models at the 3-8B scale, and if they depend on anything specific to a given modality? We answer the first with **VeloRad**, a 0.8B continuous flow-matching language model that generates reports by integrating a condition-driven velocity field over a frozen text-encoder latent space, and attains the best support-weighted clinical F1 among comparable systems while remaining balanced across precision and recall. To answer the second, we apply VeloRad to 2D chest X-ray on MIMIC-CXR, a large and independently studied RRG benchmark, by swapping the vision encoder for a 2D SigLIP model and retraining under the same architecture and training recipe used for CT, with no X-ray-specific components. The same generation behavior holds across modalities.

Another contribution of this paper is a diagnosis of *why* a flow-matching CT reporter behaves as it does, and *where* its remaining clinical-accuracy gap lives. We show that the flow-matching L2 objective is misaligned with the per-finding

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